

3.6 Cyber security

Content	Additional information
Define the term cyber security and be able to describe the main purposes of cyber security.	Students should know that cyber security consists of the processes, practices and technologies designed to protect networks, computers, programs and data from attack, damage or unauthorised access.

3.6.1 Cyber security threats

Content	Additional information
Understand and be able to explain the following cyber security threats: • social engineering techniques • malicious code • weak and default passwords • misconfigured access rights • removable media • unpatched and/or outdated software.	
Explain what penetration testing is and what it is used for.	Penetration testing is the process of attempting to gain access to resources without knowledge of usernames, passwords and other normal means of access. Students should understand the following two types of penetration testing: • when the person or team testing the system has knowledge of and possibly basic credentials for the target system, simulating an attack from inside the system (a malicious insider). • when the person or team testing the system has no knowledge of any credentials for the target system, simulating an attack from outside the system (an external attack).

3.6.1.1 Social engineering

Content	Additional information
Define the term social engineering. Describe what social engineering is and how it can be protected against. Explain the following forms of social engineering: • blagging (pretexting) • phishing • pharming • shouldering (or shoulder surfing).	Students should know that social engineering is the art of manipulating people so they give up confidential information. Blagging is the act of creating and using an invented scenario to engage a targeted victim in a manner that increases the chance the victim will divulge information or perform actions that would be unlikely in ordinary circumstances. Phishing is a technique of fraudulently obtaining private information, often using email or SMS. Pharming is a cyberattack intended to redirect a website's traffic to another, fake site. Shouldering is observing a person's private information over their shoulder eg cashpoint machine PIN numbers.

3.6.1.2 Malicious code

Content	Additional information
Define the term 'malware'. Describe what malware is and how it can be protected against. Describe the following forms of malware: • computer virus • trojan • spyware • adware.	Malware is an umbrella term used to refer to a variety of forms of hostile or intrusive software.

3.6.2 Methods to detect and prevent cyber security threats

Content	Additional information
Understand and be able to explain the following security measures: • biometric measures (particularly for mobile devices) • password systems • CAPTCHA (or similar) • using email confirmations to confirm a user's identity • automatic software updates.	

3.7 Relational databases and structured query language (SQL)

3.7.1 Relational databases

Content	Additional information
Explain the concept of a database.	
Explain the concept of a relational database.	
Understand the following important database concepts. • table • record • field • data type • primary key • foreign key. Understand that the use of a relational database facilitates the elimination of data inconsistency and data redundancy.	Note that whilst the terms entity, attribute and entity identifier are more commonly used when an abstract model of a database is being considered, the terms given here will be used for both implementations of and abstract models of databases.